



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

FIRST SEMESTER: 2026-2027
COURSE HANDOUT

Date: 6th August 2026

In addition to part I (the General Handout for all courses, appended to the Timetable), this section provides further specific details about the course.

Course No.	BITS U104
Course Title	Foundations of Measurements and Experimentation
Instructor-in-Charge	Jayendra N. Bandyopadhyay
Instructors	Raj Kumar Gupta, Biswadip Shome, Sanjay Kumar Verma, Sushmita Basu, Soumya Saxena, Shraddha Gupta, Ajaya Kumar Pani, Bibhas R. Sarkar, Preeti Meena, Susmita Pradhan, Mukesh Kumar, Ravindra Dwivedi, Saurabh Gupta, M. Nyshitha Varma, Aneesh A. M., Hemant Sharma, A. R. Harikrishnan, Jagpreet Singh Sidhu, Aditi Pandey, Jyoti, Manjuladevi V., Jayendra N. Bandyopadhyay

- 1. Course Description:** This course introduces scientific and engineering inquiry through the principles of measurement and experimentation. It covers fundamental concepts, such as accuracy, precision, calibration, error analysis, and uncertainty, as well as the hands-on use of sophisticated instruments across multiple disciplines. The course includes data acquisition, visualization, and analysis using modern computational tools. It emphasizes experimental design, hypothesis formulation, and experimental validation. It also focuses on correlating measurements with real-world problems, documenting and interpreting results, and communicating findings effectively with due consideration to ethical practices in experimentation.
- 2. Scope & Objective of the Course:** The course aims to provide a foundational understanding of measurement as an essential element of engineering and scientific practice. It covers key measurement concepts, exposure to a variety of instruments and sensors across disciplines, and methods for reliable data acquisition and analysis. The objectives of the course are to enable students to:
 - (i) Understand fundamental measurement concepts such as accuracy, precision, resolution, calibration, and uncertainty.
 - (ii) Use appropriate measuring instruments and sensors for acquiring experimental data across different domains.
 - (iii) Apply techniques for data acquisition, visualization, and analysis using modern computational tools.
 - (iv) Develop an understanding of experimental methodologies, including hypothesis formulation, experimental design, and validation.
 - (v) Interpret experimental results and correlate them with underlying scientific and engineering principles.
 - (vi) Communicate experimental findings effectively and adhere to ethical practices in experimentation and technical reporting.



3. Course Learning Outcomes: By the end of the course, students will be able to:

CO1: Understand and apply fundamental measurement concepts such as accuracy, precision, resolution, calibration, and uncertainty.

CO2: Use appropriate measuring instruments/sensors to solve a problem and acquire experimental data across different disciplines.

CO3: Analyze experimental data using graphical and statistical methods, including error estimation and uncertainty propagation.

CO4: Formulating hypotheses and their validation through design and conducting experiments.

CO5: Interpret experimental observations using fundamental scientific and engineering principles.

CO6: Document and communicate experimental findings effectively through structured reports and presentations.

4. Textbook: There is no textbook for this course. Lab manuals will be distributed to the students.

5. Reference Books:

- (i) Measurement Systems: Application and Design, Ernest O. Doebelin, McGraw-Hill, 2004, 5th Ed.
- (ii) Data Reduction and Error Analysis for the Physical Sciences, Philip R. Bevington and D. K. Robinson, McGraw-Hill, 2003, 3rd Ed.
- (iii) An Introduction to Error Analysis, John R Taylor, 2nd Ed, University Science Books, 1997.
- (iv) The Visual Display of Quantitative Information, Edward R. Tufte, Graphics Pr., 2001.

6. Course Plan:

Module No.	Lec. No.	Topics
0. Introduction	1	Overview of the Course
1. Understand fundamental measurement concepts	2-3	Accuracy, precision, resolution, calibration, and uncertainty, use of appropriate measuring instruments and sensors for acquiring experimental data across different domains, data acquisition, visualization, and analysis using modern computational tools.
2. Designing and executing experiments	4-11	Conduct the experiments pertaining to the given departments, interpret experimental results and correlate them with underlying scientific and engineering principles, and validation. (8 Experiments)



7. Evaluation Scheme:

Components	Duration	Marks (%)	Date & Time	Nature of component (Closed Book/ Open Book)
Daily Evaluation in the Labs	NA	70%	Lab Hours	Open Book
Comprehensive Examination (MCQ type)	To be announced	30%	To be announced	Closed Book

9. Attendance Policy: Each student is expected to attend all classes and participate in all evaluation components.

10. NC Criteria:

- Students have to perform 50% of the experiments
- Students have to appear for the Comprehensive Examination

11. Notices: On Nalanda

12. Make-up Policy: Stringent. Only in the most genuine cases, as judged by the IC.

13. Academic Honesty and Integrity Policy: All students must maintain academic honesty and integrity throughout the semester; no form of academic dishonesty is acceptable.

Instructor-in-charge
Course No. BITS U104